Counting and probability

Part VIII Appendix

Scope

Define sample objects and common denominators before using counting, conditional probability, Bayes' rule, and indicators. Derive geometric and binomial moments and tail bounds. Distinguish levels of independence, zero-probability and support boundaries, a centered deviation r, and a total threshold t.

Prerequisites

Permutations and combinations, probability axioms, and random variables.

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Appendix route

Learning objectives

- Count by products, disjoint sums, and complements.
- Derive expectation and variance for geometric and binomial distributions.
- Bound binomial tails using exponential moments.
- C.1 Counting
- C.2 Probability
- C.3 Discrete random variables
- C.4 The geometric and binomial distributions
- C.5 The tails of the binomial distribution

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Counting starts with the sample object

Define what one outcome records. The sum rule requires disjoint cases. The product rule requires a known branch count after each prefix. The available choices may depend on that prefix, provided each level's count is fixed. Variable branching requires summing branches rather than multiplying unspecified averages.

From six students, two distinct office holders give $6 \cdot 5 = 30$ ordered assignments. An unordered two-person committee is counted twice by those assignments, so there are 15 committees. Allowing one person to hold both offices gives $6^2 = 36$ assignments. These count three different objects.

There are q^m length-m strings over q symbols, retaining positions and allowing repetition. Substrings are contiguous; subsequences may skip positions. Specify whether identical contents at distinct positions count separately. For an n-character string and $k \geq 1$, the number of positional length-k substrings is $\max(0, n - k + 1)$. This need not equal the number of distinct contents.

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Permutations, combinations, and uniform overcounting

There are $n!$ full permutations of n distinct records. Ordered selection of k records without replacement has

$$P(n,k) = \frac{n!}{(n-k)!}, \qquad 0 \le k \le n.$$

Every unordered k-subset has exactly $k!$ orderings, giving

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

The case $k = 0$ gives one empty choice. Extend the coefficient to 0 for $k < 0$ or $k > n$.

Distinct record identities still give $n!$ permutations when keys repeat. If only key strings matter and the multiplicities are $m_1, \ldots, m_s$, each string has exactly $\prod_{i=1}^{s} m_i!$ identity permutations. Thus the number of distinct strings is $n! / \prod_{i=1}^{s} m_i!$. Prove equal overcounting before dividing.

Complementation gives $\binom{n}{k} = \binom{n}{n-k}$. Partitioning subsets by whether they include a specified element gives Pascal's identity:

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

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Binomial expansion and entropy bounds

In $(x+y)^n$, there are $\binom{n}{k}$ ways to choose the k factor positions contributing x. Hence

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Setting $x = y = 1$ gives $\sum_{k=0}^{n} \binom{n}{k} = 2^n$, also counting subsets by size. For $1 \le k \le n$,

$$\binom{n}{k} = \prod_{j=0}^{k-1} \frac{n-j}{k-j} \ge \left(\frac{n}{k}\right)^k,$$

because each factor is at least n/k. Also,

$$\binom{n}{k} \le \frac{n^k}{k!} \le \left(\frac{en}{k}\right)^k,$$

using $k! \ge (k/e)^k$.

Let $\alpha = k/n \in (0,1)$. The nonnegative kth term in the expansion of $(\alpha + 1 - \alpha)^n = 1$ is at most 1. Therefore

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Binomial expansion and entropy bounds (continued)

$$\binom{n}{k} \leq \alpha^{-k}(1-\alpha)^{-(n-k)} = 2^{nH(\alpha)},$$

where $H(\alpha) = -\alpha \log_2 \alpha - (1-\alpha)\log_2(1-\alpha)$. Extend to the endpoints with $0 \log_2 0 = 0$, giving coefficient 1. Since $H'(\alpha) = \log_2((1-\alpha)/\alpha)$, the maximum is $H(1/2) = 1$.

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Probability axioms and uniform outcomes

An elementary outcome records one complete distinguishable result. An event is a measurable subset of the sample space S. On the finite or countable discrete spaces here, every subset is an event. An uncountable space requires a specified event family closed under complements and countable unions; arbitrary subsets are not automatically measurable.

Probability is nonnegative, satisfies $P(S) = 1$, and is finitely or countably additive on pairwise disjoint events. It follows that

$$P(\varnothing) = 0, \qquad P(A^c) = 1 - P(A), \qquad A \subseteq B \implies P(A) \le P(B).$$

In a discrete space, $P(A) = \sum_{s \in A} P(\{s\})$.

The formula $P(A) = |A|/|S|$ requires finitely many equally likely elementary outcomes. Two independent fair bits have four ordered outcomes, each with probability $1/4$. Compressing them to the number of ones gives values $0, 1, 2$ with probabilities $1/4, 1/2, 1/4$, respectively. Those three values are not uniform.

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Union bounds need no independence

Disjoint decomposition into the A-only part, the intersection, and the B-only part gives

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B).$$

For many events, define

$$B_1 = A_1, \qquad B_i = A_i \setminus \bigcup_{j<i} A_j.$$

The B_i are disjoint, have the same union as the A_i, and satisfy $B_i \subseteq A_i$. Additivity and monotonicity give

$$P\left(\bigcup_i A_i\right) = \sum_i P(B_i) \leq \sum_i P(A_i).$$

This also holds for countably many events and needs no independence. If the right side exceeds 1, cap it at 1. By contrast, factoring a general collection of intersection probabilities requires mutual independence.

Union bounds need no independence (continued)

If three failure events each have probability at most 0.01, their union has probability at most 0.03, regardless of dependence. If the three events are identical and each has probability 0.01, their union has probability 0.01. The expression $1 - 0.99^3$ requires independence.

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Conditional probability and the chain rule

For $P(B) > 0$, conditional probability renormalizes the outcomes retained in B:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(B)P(A \mid B).$$

Repeated application yields

$$P\left(\bigcap_{i=1}^{n} A_i\right) = P(A_1) \prod_{i=2}^{n} P\left(A_i \,\middle|\, \bigcap_{j=1}^{i-1} A_j\right).$$

Every conditioning prefix must have positive probability. A zero-probability prefix already makes the total intersection zero; subsequent undefined ratios must not be multiplied.

A bag contains three blue and two red balls. Drawing twice without replacement gives $P(\text{both blue}) = (3/5)(2/4) = 3/10$. The second draw's blue marginal remains $3/5$, but conditional on a first blue it is $1/2$. Thus the draws are dependent. Replacement with fresh uniform selection gives $(3/5)^2$ instead.

Conditional probability and the chain rule (continued)

Knowing that at least one occurrence happened usually differs from knowing that a specified occurrence happened. Observation and selection rules belong to the model. Two remaining verbal cases need not each have probability $1/2$.

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Independence, disjointness, and pairwise independence

Events A, B are independent when $P(A \cap B) = P(A)P(B)$. This includes probability-zero events. The equivalent expression $P(A \mid B) = P(A)$ additionally needs $P(B) > 0$. Disjoint events with positive probabilities are dependent. The empty event is independent of every event.

Mutual independence requires the intersection probability of every finite subcollection to factor into the product of its marginals. Pairwise independence checks only pairs and cannot justify factoring three or more events.

Let U, V be independent fair bits, and define $W = U \operatorname{XOR} V$. The outcomes $000, 011, 101, 110$ each have probability $1/4$. Each of the pairs (U, V), (U, W), and (V, W) has all four combinations equally likely, so every pair is independent. Yet

$$P(U = V = W = 1) = 0 \neq \frac{1}{8} = P(U = 1)P(V = 1)P(W = 1).$$

The three bits are not mutually independent.

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Independence, disjointness, and pairwise independence (continued)

Pairwise independent, but not mutually independent

U	V	W=U XOR V	P(outcome)
0	0	0	1/4
0	1	1	1/4
1	0	1	1/4
1	1	0	1/4

Each pair:00/01/10/11, each1/4

All three P(=1) marginals are1/2

But111 is absent from sample space

$0 \neq (1/2)^3 = 1/8$

Pair checks do not factor triple intersections. Variances add here; triple products do not.

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Total probability and Bayes denominators

Let the B_i form a countable partition of S, with $P(B_i) > 0$. The event A is the disjoint union of $A \cap B_i$, so

$$P(A) = \sum_i P(B_i) P(A \mid B_i).$$

When $P(A) > 0$, Bayes' rule gives

$$P(B_j \mid A) = \frac{P(B_j) P(A \mid B_j)}{P(A)}.$$

A monitor has fault prior 0.02, sensitivity 0.9, and false-positive rate 0.1. Its alarm probability and posterior fault probability are

$$P(\text{alarm}) = 0.02 \cdot 0.9 + 0.98 \cdot 0.1 = 0.116,$$

$$P(\text{fault} \mid \text{alarm}) = \frac{0.018}{0.116} = \frac{9}{58} \approx 0.155172.$$

Sensitivity 0.9 is not posterior probability 0.9. Normal cases contribute 0.098 false alarms to the denominator. If two tests are independent conditional on fault status, then

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Total probability and Bayes denominators (continued)

$$P(\text{two alarms}) = 0.02 \cdot 0.9^2 + 0.98 \cdot 0.1^2 = 0.026,$$

$$P(\text{fault} \mid \text{two alarms}) = \frac{0.0162}{0.026} = \frac{81}{130} \approx 0.623077.$$

Squaring the likelihoods requires that conditional independence assumption.

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Continuous uniform probabilities

Let X be continuous uniform on $[a, b]$, where $a < b$. For $a \le c \le d \le b$,

$$P(c \le X \le d) = \frac{d - c}{b - a}.$$

Each singleton has probability 0, while the whole interval has probability 1. Countable additivity does not call for summing uncountably many singleton masses.

On $[2, 8]$, $P(3 \le X \le 5) = 2/6 = 1/3$, and removing endpoints changes nothing. The variable $Z = \lfloor X \rfloor$ has values $2, \ldots, 7$, each with probability $1/6$. The value $Z = 8$ occurs only when $X = 8$, with probability 0. Do not assign equal weights to seven integer labels.

The following sections concern discrete random variables and probability mass. A continuous density may exceed 1 even though every singleton has probability zero. Integrate a density over intervals rather than treating it as point probability.

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Discrete random variables and joint masses

A random variable $X : S \to \mathbb{R}$ assigns a numerical value to an outcome. The event $\{X = x\}$ is a preimage. Its probability mass function is $p_X(x) = P(X = x)$, with nonnegative values satisfying $\sum_x p_X(x) = 1$. The word “mass” distinguishes discrete probabilities from continuous density.

The joint mass $p(x, y) = P(X = x, Y = y)$ gives marginals

$$p_X(x) = \sum_y p(x, y), \qquad p_Y(y) = \sum_x p(x, y).$$

For $p_Y(y) > 0$, the conditional mass is $p(x, y)/p_Y(y)$. Independence means $p(x, y) = p_X(x)p_Y(y)$ for every pair (x, y).

Let A, B be independent fair four-sided dice, and let $M = \max(A, B)$. Then

$$P(M \le m) = \frac{m^2}{16}, \qquad P(M = m) = \frac{m^2 - (m-1)^2}{16} = \frac{2m - 1}{16}.$$

For $m = 1, 2, 3, 4$, the masses are $1, 3, 5, 7$ divided by 16. They are not uniform.

Expectation need not be a possible outcome

Expectation is $E[X] = \sum_x x p_X(x)$. Finite support always permits evaluation. On countable signed support, $E[|X|] < \infty$ ensures a finite expectation that may be reordered. A nonnegative X may have infinite expectation. The expression $\infty - \infty$ is not valid arithmetic.

For the dice maximum,

$$E[M] = \frac{1 + 6 + 15 + 28}{16} = \frac{50}{16} = \frac{25}{8},$$
$$E[M^2] = \frac{1 + 12 + 45 + 112}{16} = \frac{170}{16} = \frac{85}{8}.$$

The mean 3.125 is outside the integer support and is still correct.

For any g satisfying $E[|g(X)|] < \infty$, grouping outcomes by their X-preimages gives

$$E[g(X)] = \sum_x g(x) p_X(x).$$

There is no need to list the distinct values of $g(X)$ first. In general, $E[g(X)] \neq g(E[X])$; for the square function, the difference is variance.

Linearity and indicators

For integrable X, Y, expand using their joint mass:

$$\begin{aligned} E[X+Y] &= \sum_x \sum_y (x+y)p(x,y) \\ &= \sum_x x \sum_y p(x,y) + \sum_y y \sum_x p(x,y) \\ &= E[X] + E[Y]. \end{aligned}$$

Absolute integrability justifies splitting and reordering. Independence was not used. Also $E[aX + b] = aE[X] + b$, and repeated application gives linearity for finite sums.

The indicator I_A equals 1 on A and 0 otherwise, so $E[I_A] = P(A)$. If X counts events, then $X = \sum_i I_i$ gives $E[X] = \sum_i P(A_i)$, even with dependence. In a uniform permutation of n distinct keys, each inversion-pair indicator has expectation $1/2$, giving total expected inversions $n(n-1)/4$.

For independent integrable X, Y, joint factorization gives

Linearity and indicators (continued)

$$E[XY] = \left(\sum_x x p_X(x)\right)\left(\sum_y y p_Y(y)\right) = E[X]E[Y].$$

This step uses independence. If $X = Y$ is one fair bit, $E[XY] = 1/2 \neq 1/4$. Products of many variables generally require mutual independence, rather than pairwise independence alone.

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Tail sums and Jensen

A nonnegative integer variable satisfies, outcome by outcome,

$$X = \sum_{i \geq 1} I_{\{X \geq i\}}.$$

Take expectations of finite truncations, then use monotone limits of nonnegative terms. This gives

$$E[X] = \sum_{i \geq 1} P(X \geq i),$$

possibly with value ∞. A signed variable cannot use this formula directly.

A convex function satisfies

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y), \qquad 0 \leq \lambda \leq 1.$$

For a finite support $x_1, \ldots, x_m$ with weights w_i, let $s = \sum_{i<m} w_i$. When $0 < s < 1$, combine the first $m-1$ points into $z = \sum_{i<m} w_i x_i / s$. Two-point convexity and induction give

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Tail sums and Jensen (continued)

$$\begin{aligned} f\left(\sum_{i=1}^{m} w_i x_i\right) &= f(sz + w_m x_m) \\ &\leq s f(z) + w_m f(x_m) \\ &\leq \sum_{i=1}^{m} w_i f(x_i). \end{aligned}$$

Zero weights may be omitted, and the single-point case is equality. Thus $f(E[X]) \leq E[f(X)]$.

For the general integrable case, assume f is convex on an interval containing the support and $\mu = E[X]$. A supporting line at an interior μ has

$$f(x) \geq f(\mu) + a(x - \mu).$$

Taking expectations gives $E[f(X)] \geq f(\mu) + a(E[X] - \mu) = f(\mu)$. At a degenerate endpoint, X is concentrated there. The relevant expectations must be defined. Jensen is not permission to move an arbitrary function through expectation.

Variance, covariance, and dependence

For $E[X^2] < \infty$ and $\mu = E[X]$,

$$\operatorname{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2 \geq 0.$$

The standard deviation is $\sigma = \sqrt{\operatorname{Var}(X)}$. Scaling gives $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$. For the dice maximum,

$$\operatorname{Var}(M) = \frac{85}{8} - \left(\frac{25}{8}\right)^2 = \frac{55}{64}, \qquad \sigma = \frac{\sqrt{55}}{8}.$$

Define $\operatorname{Cov}(X,Y) = E[XY] - E[X]E[Y]$. Expanding squares gives

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y),$$

$$\operatorname{Var}\left(\sum_{i} X_i\right) = \sum_{i} \operatorname{Var}(X_i) + 2\sum_{i<j} \operatorname{Cov}(X_i, X_j).$$

Pairwise independence and finite second moments suffice to make every covariance zero. Mutual independence is unnecessary for variance addition.

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Variance, covariance, and dependence (continued)

Identical marginals alone do not suffice. If all n variables are the same fair bit, their sum equals 0 or n with equal probability and has variance $n^2/4$. Adding individual variances would give only $n/4$. Omitting covariance underestimates uncertainty.

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Markov and Chebyshev assumptions

Let $X \geq 0$ have finite mean, and let $t > 0$. The pointwise inequality

$$X \geq tI_{\{X \geq t\}}$$

gives $E[X] \geq tP(X \geq t)$ after taking expectations. Thus Markov's inequality is

$$P(X \geq t) \leq \frac{E[X]}{t}.$$

It requires no independence. It cannot be applied directly to signed variables, where positive and negative cancellation may hide a large positive tail.

For Y with finite second moment, set $X = (Y - \mu)^2$ and use the threshold a^2, where $a > 0$. Markov gives Chebyshev's inequality:

$$P(|Y - \mu| \geq a) = P((Y - \mu)^2 \geq a^2) \leq \frac{\mathrm{Var}(Y)}{a^2}.$$

This is a two-sided bound. It also bounds either one-sided event, although it may be loose.

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Markov and Chebyshev assumptions (continued)

For the maximum M, use $\mu = 25/8$, variance $55/64$, and $a = 2$. The bound is $55/256 \approx 0.214844$. The actual event contains only $M = 1$, with probability $1/16$. A bound is not an equality; values above 1 may be capped at 1.

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Geometric and binomial distributions

A Bernoulli variable with parameter p equals 1 with probability p and 0 with probability $q = 1 - p$. Since $X^2 = X$, it has $E[X] = p$ and $\operatorname{Var}(X) = p - p^2 = pq$. The trial sequences below additionally assume mutual independence and a common p. Equal marginals alone are insufficient.

Here the geometric variable G counts trials including the first success. Its support is $1, 2, \ldots$, with $0 < p \leq 1$. The event $G = k$ requires $k - 1$ failures followed by success, so

$$P(G = k) = q^{k-1}p.$$

At $p = 0$, success never arrives and $G = \infty$, outside this finite-valued distribution.

Counting earlier failures gives $F = G - 1$, with support $0, 1, \ldots$, mass $P(F = f) = q^f p$, and mean q/p. Do not use the trial mean $1/p$ for both conventions. At $p = 1$, $G \equiv 1$, $F \equiv 0$, and both variances are zero.

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Geometric normalization and moments

The geometric masses normalize because

$$\sum_{k\ge 1} q^{k-1}p = \frac{p}{1-q} = 1.$$

For an integer $t \ge 0$, $P(G > t) = q^t$, hence $P(G \ge i) = q^{i-1}$. Tail summation gives

$$E[G] = \sum_{i\ge 1} q^{i-1} = \frac{1}{1-q} = \frac{1}{p}.$$

The weighted geometric identity yields

$$\sum_{k\ge 1} k^2 q^{k-1} = \frac{1+q}{(1-q)^3}, \qquad E[G^2] = \frac{1+q}{p^2}.$$

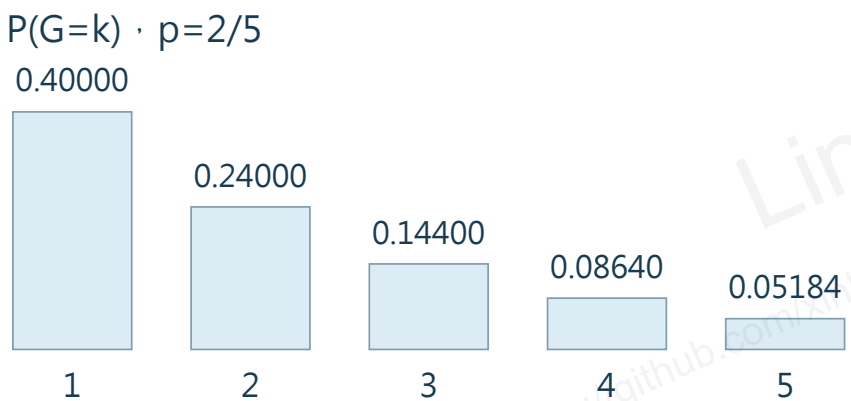
Subtracting the squared mean gives $\operatorname{Var}(G) = q/p^2$. Handle $q = 0$ directly as $G \equiv 1$, without dividing by q.

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Geometric normalization and moments (continued)

At $p = 2/5$, the first five masses are $2/5$, $6/25$, $18/125$, $54/625$, and $162/3125$. Their sum is $2882/3125$, and the remaining probability is $P(G > 5) = 243/3125$. The mean is $5/2$ and the variance is $15/4$. A mean of 2.5 averages repeated complete experiments; it does not place a success on a fractional trial.

Fix the counting convention before masses and tails



One outcome: F → F → F → S

G=4: includes the success trial

F=G−1=3: failures only

No success in five: $q^5 = 243/3125$

This tail remains beyond the first five bars

Axis counts trials including first success. E[G]=2.5,E[F]=1.5; the counts differ.

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Memorylessness and retry-model boundaries

For $0 < p < 1$ and integers $s, t \geq 0$,

$$P(G > s + t \mid G > s) = \frac{q^{s+t}}{q^s} = q^t = P(G > t).$$

Past failures do not change the next success rate because the trials are independent with a common p. A long failure streak does not make success due.

Changing retry probabilities, persistent shared faults, and sampling without replacement violate that model. Independent trials with unequal p_i instead have tail $\prod_{i=1}^{t}(1 - p_i)$. Dependence may require conditional probabilities rather than q^t.

A fixed trial cost c gives expected total cost c/p. Different costs for failure and success give

$$\text{total cost} = c_{\text{fail}}(G - 1) + c_{\text{success}},$$
$$E[\text{total cost}] = c_{\text{fail}}\frac{q}{p} + c_{\text{success}}.$$

Specify how costs relate to trials before applying linearity.

Binomial mass, mean, and variance

For n mutually independent trials with a common success probability p, let $X = \sum_{i=1}^{n} I_i \in \{0, \ldots, n\}$. A specified set of k success positions has probability $p^k q^{n-k}$. There are $\binom{n}{k}$ disjoint such position sets, so

$$b(k; n, p) = \binom{n}{k} p^k q^{n-k}.$$

Normalization follows from $(p + q)^n = 1$.

Linearity gives $\mu = np$ without independence. Variance addition requires zero pairwise covariances, giving $\operatorname{Var}(X) = npq$. The complete binomial mass formula used mutual independence. Pairwise independence suffices for the mean and variance but does not imply the distribution.

At $n = 0$ or $p = 0$, X is concentrated at 0; at $p = 1$, it is concentrated at n. The endpoint convention $0^0 = 1$ represents an empty product.

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Binomial mass, mean, and variance (continued)

Mutually independent trials with unequal p_i instead have a Poisson-binomial distribution, with mean $\sum_i p_i$ and variance $\sum_i p_i(1 - p_i)$. Substituting the average p into a binomial mass formula does not make the distributions equal.

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A fully checkable binomial table

For $n = 8, p = 1/4, q = 3/4$, the common denominator is $4^8 = 65536$. The numerators are $\binom{8}{k} 3^{8-k}$.

k	0	1	2	3	4	5	6	7	8
Numerator	6561	17496	20412	13608	5670	1512	252	24	1

The numerators sum to 65536. Their k-weighted sum is 131072, giving $E[X] = 2$. Their k^2-weighted sum is 360448, giving $E[X^2] = 11/2$ and variance $11/2 - 4 = 3/2$.

The exact tail is

$$P(X \geq 5) = \frac{1512 + 252 + 24 + 1}{65536} = \frac{1789}{65536} \approx 0.027298.$$

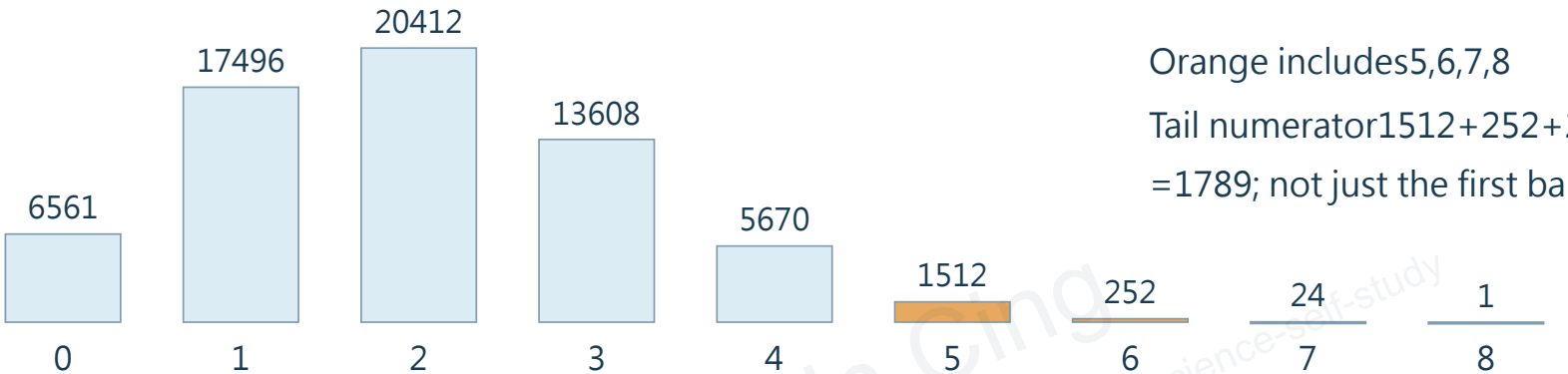
It includes every term through $k = 8$, rather than only $b(5)$.

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A fully checkable binomial table (continued)

A single mass differs from a tail; threshold5 includes5

Bar height=numerator/65536; n8,p1/4



Orange includes5,6,7,8

Tail numerator1512+252+24+1

=1789; not just the first bar

Numerators sum65536; mean2,variance1.5; P(X≥5)=1789/65536≈.027298.

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Modes follow adjacent ratios

For $0 < p < 1$ and $1 \le k \le n$,

$$\frac{b(k)}{b(k-1)} = \frac{(n-k+1)p}{kq}.$$

The ratio exceeds 1 exactly when $k < (n+1)p$, and equals 1 when $k = (n+1)p$.

If $(n+1)p$ is not an integer, the unique mode is $\lfloor (n+1)p \rfloor$. If it is an integer m, both $m-1$ and m are modes. For $n = 8, p = 1/4$, the mode is 2 and coincidentally equals the mean 2. For $n = 4, p = 1/2$, $(n+1)p = 2.5$ gives mode 2. For $n = 3, p = 1/2$, the modes are $1, 2$, while the mean 1.5 has no integer probability bar.

The coefficient and entropy bound gives

$$b(k) \le \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}.$$

Interpret endpoints by continuous extension or the 0^0 convention. This bounds one mass, rather than a complete tail.

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Binomial tails and union bounds

For an integer-valued X, $P(X \geq a)$ sums from $\lceil a \rceil$, whereas $P(X > a)$ starts at $\lfloor a \rfloor + 1$. For an integer k, $P(X < k)$ ends at $k - 1$, unlike $P(X \leq k)$.

For $X \sim \mathrm{Bin}(n, p)$, the event $X \geq k$ implies that some specified set of k trials all succeed. Each such set has probability p^k. Applying the union bound over the $\binom{n}{k}$ sets gives

$$P(X \geq k) \leq \binom{n}{k} p^k, \qquad 0 \leq k \leq n.$$

The events need not be disjoint. At $k = 0$, the bound is 1; at $k > n$, the actual event is empty. Apply the same argument to failures $Y = n - X \sim \mathrm{Bin}(n, q)$ to get

$$P(X \leq k) \leq \binom{n}{k} q^{n-k}, \qquad 0 \leq k \leq n.$$

At $n = 8, p = 1/4, k = 5$, the upper bound is $56/1024 = 7/128 \approx 0.054688$, compared with the exact tail $1789/65536 \approx 0.027298$. Outcomes with six or more successes are counted repeatedly, which explains why the bound is usually strict.

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Deriving a geometric left-tail ratio bound

Assume $0 < p < 1$ and an integer $0 < k < np$. For $1 \le i \le k$,

$$\frac{b(i-1)}{b(i)} = \frac{iq}{(n-i+1)p} < \frac{kq}{(n-k)p} = \rho < 1.$$

Working backward gives $b(k-j) < \rho^j b(k)$ for $j = 1, \dots, k$. Consequently,

$$\begin{aligned} P(X < k) &= \sum_{i=0}^{k-1} b(i) < b(k) \sum_{j \ge 1} \rho^j \\ &= \frac{\rho}{1-\rho} b(k) = \frac{kq}{np-k} b(k). \end{aligned}$$

The denominator is positive by assumption. Substituting $k = np$ is invalid.

If $0 < k \le np/2$, the coefficient is at most $q \le 1$, so $L_k = P(X < k) < b(k)$. Since $L_{k+1} = L_k + b(k)$,

$$\frac{L_k}{L_{k+1}} = \frac{L_k}{L_k + b(k)} < \frac{1}{2}.$$

Moving one integer threshold farther into the left tail at least halves its probability.

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Right tails change both variables and thresholds

Let $Y = n - X \sim \mathrm{Bin}(n, q)$. For an integer $np < k < n$, we have $n - k < nq$. Apply the left-tail bound at threshold $n - k$:

$$\begin{aligned} P(X > k) &= P(Y < n - k) \\ &< \frac{(n-k)p}{nq - (n-k)} b(n-k; n, q) \\ &= \frac{(n-k)p}{k - np} b(k; n, p). \end{aligned}$$

If $k > (np + n)/2$, the left-tail halving result gives

$$P(X > k) < \frac{1}{2} P(X > k - 1).$$

This exchanges the roles of success and failure, including their thresholds.

At $n = 8, p = 1/4, k = 4$, the coefficient is

$$\frac{(8-4)(1/4)}{4-2} = \frac{1}{2}.$$

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Right tails change both variables and thresholds (continued)

Thus $P(X > 4) < 2835/65536 \approx 0.043259$, compared with the exact tail $1789/65536$. This improves the union bound in this example, but need not do so for every parameter choice.

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Exponential moments turn tails into nonnegative quantities

Let I_i be mutually independent Bernoulli indicators with parameters p_i. Write $X = \sum_i I_i$ and $\mu = \sum_i p_i$. For $\lambda > 0$, monotonicity and Markov give

$$P(X \geq t) \leq e^{-\lambda t} E[e^{\lambda X}].$$

Independence permits factorization. Using $1 + z \leq e^z$,

$$\begin{aligned} E[e^{\lambda X}] &= \prod_i E[e^{\lambda I_i}] \\ &= \prod_i [1 + p_i(e^\lambda - 1)] \\ &\leq \exp[\mu(e^\lambda - 1)]. \end{aligned}$$

Therefore $P(X \geq t) \leq \exp[\mu(e^\lambda - 1) - \lambda t]$. For $\mu > 0, t > \mu$, differentiating the exponent gives $\mu e^\lambda - t = 0$. The minimum occurs at $\lambda = \ln(t/\mu) > 0$, because the second derivative is $\mu e^\lambda > 0$. Substitution gives

$$P(X \geq t) \leq e^{-\mu} \left(\frac{e\mu}{t} \right)^t.$$

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Exponential moments turn tails into nonnegative quantities (continued)

Unequal p_i are allowed; mutual independence is required. If $\mu = 0$, every indicator is zero almost surely. All positive-threshold tails are zero directly, without evaluating $\ln(t/0)$.

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Centered r and total t are different thresholds

The looser CLRS form is

$$P(X - \mu \geq r) \leq \left(\frac{e\mu}{r}\right)^r, \qquad r > \mu > 0.$$

To derive it directly, for $\lambda > 0$ bound each centered exponential moment:

$$\begin{aligned} E[e^{\lambda(I_i - p_i)}] &= p_i e^{\lambda(1-p_i)} + (1-p_i)e^{-\lambda p_i} \\ &\leq p_i e^{\lambda} + 1 \leq \exp(p_i e^{\lambda}). \end{aligned}$$

Factor the moments using mutual independence and apply Markov:

$$\begin{aligned} P(X - \mu \geq r) &\leq e^{-\lambda r} \prod_i E[e^{\lambda(I_i - p_i)}] \\ &\leq \exp(\mu e^{\lambda} - \lambda r). \end{aligned}$$

The exponent is minimized at $\lambda = \ln(r/\mu) > 0$, giving the stated bound.

Here r is the deviation above the mean, so the total threshold is $t = \mu + r$. In the preceding formula, t counts total successes. With $\mu = 10, r = 40$, the event is $X \geq 50$. The two bounds are

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Centered r and total t are different thresholds (continued)

$$\left(\frac{e}{4}\right)^{40} \approx 1.947 \times 10^{-7},$$

$$e^{-10}\left(\frac{e}{5}\right)^{50} \approx 2.650 \times 10^{-18}.$$

Exchanging 40 and 50 changes the derivation. When r only slightly exceeds μ, the centered bound may exceed 1; cap it at 1. For an integer binomial variable, the exact tail starts at $\lceil \mu + r \rceil$, never a rounded-down threshold.

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Compare guarantees on one binomial tail

Let $X \sim \mathrm{Bin}(1000, 0.01)$, so $\mu = 10$, and consider $X \geq 50$.

Bound	Calculation	Upper bound
Markov	$10/50$	0.2
Chebyshev	$9.9/40^2$	0.0061875
Centered, $r = 40$	$(e/4)^{40}$	$\approx 1.947 \times 10^{-7}$
Total threshold, $t = 50$	$e^{-10}(e/5)^{50}$	$\approx 2.650 \times 10^{-18}$

The exact value remains

$$\sum_{k=50}^{1000} \binom{1000}{k} 0.01^k 0.99^{1000-k}.$$

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Compare guarantees on one binomial tail (continued)

An adjacent-ratio recurrence or log masses avoid floating-point overflow from large factorials. Markov requires nonnegativity and the mean. Chebyshev requires the true variance, which is 9.9 here. Both exponential bounds require mutually independent indicators.

If all indicators instead share the same Bernoulli variable with parameter 0.01, then X is 0 or 1000, and $P(X \geq 50) = 0.01$. Markov still applies, but the binomial variance and exponential-product factorization do not. Equal marginals do not justify the binomial or Chernoff model.

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Sources

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References and attribution

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